Seasonal Recharge of a Shallow Alluvial Aquifer

R. Okonjo and L. Varga

1. Introduction

Shallow alluvial aquifers supply most of the drinking water in the villages of the lower valley. Their recharge depends on a short rainy season and on infiltration from the riverbed.[1] Little is known about how quickly the water table responds after the first heavy rain.

2. Method

We monitored twelve wells twice a week from May to November. The decline of the water level after the last rain was fitted with the exponential model $h(t) = h_0 \cdot \mathrm{e}^{-kt}$, where k is the recession constant. Rainfall was recorded at a station located 3 km from the wells.[2]

3. Results

The water table rose by an average of 2.4 m within ten days of the first storm. The recession constant was remarkably similar across all twelve wells. These results suggest that the aquifer is recharged mainly through the riverbed rather than by direct infiltration.

[1] Infiltration was estimated from double-ring tests at six points.

[2] The station is operated by the district agricultural office.